DSA210 PROPOSAL REPORT

Introduction:

The COVID-19 pandemic caused unprecedented disruptions to global education systems. This project aims to quantify the "learning loss" by comparing standardized exam performances across three distinct phases: **Pre-COVID (2017–2019)**, **Lockdown/Transition (2020–2022)**, and **Recovery (2023–2025)**. The goal is to determine if academic performance has returned to pre-pandemic trajectories or if a "permanent" shift in performance has occurred.

Data Source & Collection:

The primary dataset will be obtained from the Stanford Education Data Archive (SEDA), which aggregates district-level academic performance for public schools in the United States from 2009 to the present. I will extract data for the years 2017–2025 (pre-COVID: 2017–2019; during-COVID: 2020–2022; post-COVID: 2023–2025). The data is publicly available for download in CSV format, and I will use python's tools and libraries to load and filter the required years.

Data Characteristics:

The dataset includes approximately 15000 - 20000 records, each with:

- District identifier, state, locale (urban, suburban, etc.)
- Average math and reading scores (mean and standard deviation)
- Demographic breakdowns (percent free/reduced-price lunch, percent English learners, racial/ethnic composition)
- Sample sizes (number of students tested per district)

Data Enrichment:

To capture the intensity of COVID-19 disruptions, I will enrich the district-level data with the **Oxford COVID-19 Government Response Tracker (OxCGRT)** stringency index. This index (0–100) quantifies the strictness of government policies, including school closures, at the state level. I will merge it by state and month, then aggregate to an annual “average stringency” for each state and year. This enrichment allows me to model the effect of varying lockdown intensity across states rather than relying on a simple pre/post-comparison.

Planned Analysis:

I will implement an **Interrupted Time Series (ITS)** analysis. I will use **Linear Regression** and **Random Forest** to model the pre-COVID trend and forecast a "Counterfactual 2020–2025" (what would have happened without the pandemic). The difference between the forecast and actual results will define the impact. **K-Means Clustering** will then be used to categorize regions into "Resilient," "Recovering," or "Struggling" groups based on their recovery slope.

Tools & Libraries:

- Pandas – Data cleaning, merging and manipulation
- Numpy – Numerical operations and handling
- Matplotlib & seaborn – Data visualizations
- Scipy.stats – For t-tests, ANOVA and other statistical tests
- Scikit-learn – Primary Machine Learning Library
- Jupyter Notebook – Interactive Development Environment + Documentation